

Perishables Management System (PMS)

Test Case Document · Version 1

Project	Perishables Management System (PMS), Software Engineering, 5th semester, B.Tech CSE (AI & ML), PES University
Based on	SRS Version 1 and System Architecture Version 1
Document version	1.0 · 5 October 2026
Coverage	83 test cases covering all 44 requirements in SRS v1 (55 positive, 22 negative, 6 boundary)

1. Introduction

Purpose. This document defines how the PMS will be tested and lists the test cases that verify each requirement in SRS Version 1.

Scope. All functional and non-functional requirements in SRS v1. Sensor and POS inputs are tested with the simulators described in the SRS.

2. Test Plan

2.1 Test levels

Level	What is tested	Tool
Unit	Functions: FEFO deduction, expiry and projection calculations, validation	pytest
Integration	API + database; simulator → device API → alert	pytest + httpx, Postman
System	End-to-end flows through the web app	Manual, Playwright
Performance	Response times and load (NFR-1, 3, 5, 20)	Locust
Security	Roles, lockout, hashing, device keys	Manual, Postman
Usability	New-user tasks (NFR-10)	Observed sessions

2.2 Test environment

- Backend (FastAPI), PostgreSQL and web app running in Docker on a Linux machine
- Chrome and Firefox on desktop; one Android phone
- POS simulator and sensor simulator scripts; sample CSV files
- Test database seeded with known products, batches and users

2.3 Entry and exit criteria

Entry: feature built and merged to the main branch; test data loaded; simulators running.

Exit: all High-priority test cases passed; no open Critical or Major defects; summary report completed.

2.4 Test case types

Positive valid input, expected behaviour **Negative** invalid input or unauthorised action
Boundary values at or around a limit

2.5 Defect severity

Severity	Meaning
Critical	Wrong stock, data loss, security breach, or system down
Major	Feature does not work; no workaround
Minor	Feature works with a workaround or small error
Cosmetic	Layout, text or colour issue

3. Test Cases

1. Authentication and Access (14 test cases)

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-AUTH-01	FR-1	Positive	Valid login	Active user staff1 (Inventory Staff) exists	1. Open login page 2. Enter staff1 and correct password 3. Click Login	Login succeeds; Inventory Staff home screen is shown	
TC-AUTH-02	FR-1	Negative	Wrong password	User staff1 exists	1. Enter staff1 with a wrong password 2. Click Login	Error "Invalid user ID or password"; user stays on login page	
TC-AUTH-03	FR-1	Negative	Unknown user ID	No user ghost99	1. Enter ghost99 with any password 2. Click Login	Same generic error as TC-AUTH-02 (does not reveal which field is wrong)	
TC-AUTH-04	FR-2	Positive	Cashier sees only cashier screens	User cash1 (Cashier)	1. Log in as cash1 2. Open the menu	Only sales screens visible; Users, Settings, Reports not shown	
TC-AUTH-05	FR-2	Negative	Direct URL / API access blocked by role	Logged in as cash1	1. Call GET /api/users directly (browser or Postman)	HTTP 403 Forbidden; attempt written to log	
TC-AUTH-06	FR-2	Positive	Manager-only action hidden from staff	Logged in as staff1	1. Open markdown rules screen 2. Call POST /api/markdown-rules	Edit controls hidden; API returns 403	
TC-AUTH-07	FR-3	Boundary	4 failures then success	User staff1, counter 0	1. Enter wrong password 4 times 2. Enter correct password	Login succeeds; failed-attempt counter resets to 0	
TC-AUTH-08	FR-3	Negative	Lockout after 5 failures	User staff1, counter 0	1. Enter wrong password 5 times 2. Enter correct password	Account locked; correct password rejected with "Account locked, contact Admin"	
TC-AUTH-09	FR-3	Positive	Idle session logout	Idle timeout set to 1 min for test	1. Log in 2. Leave the app idle for 1 min 3. Click any menu item	Session ended; user redirected to login page	
TC-AUTH-10	FR-31	Positive	Admin creates user	Logged in as Admin	1. Open Users 2. Add user cash2, role Cashier 3. Save 4. Log in as cash2	User created; cash2 logs in and sees Cashier screens	

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-AUTH-11	FR-31	Negative	Duplicate user ID	User cash2 exists	1. Add another user with ID cash2 2. Save	Validation error "User ID already exists"; not saved	
TC-AUTH-12	FR-31	Positive	Admin deactivates user	User cash2 active	1. Deactivate cash2 2. Try logging in as cash2	Login refused; cash2's past records remain in the system	
TC-AUTH-13	FR-32	Positive	Password reset by email	staff1 has a registered email	1. Click "Forgot password" 2. Enter staff1 3. Open emailed link 4. Set new password 5. Log in with new password	Reset email received; new password works; old password rejected	
TC-AUTH-14	FR-32	Negative	Reset link reused	Reset link already used once	1. Open the same reset link again	Link rejected as invalid/expired	

2. Product Master Data (5 test cases)

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-MD-01	FR-33	Positive	Add product	Logged in as Manager	1. Add Tomato: Vegetables, kg, shelf-life 5 days, chilled, 8-12 °C, cost 20, price 30, threshold 10 2. Save	Product saved and listed with all fields	
TC-MD-02	FR-33	Negative	Missing mandatory field	Logged in as Manager	1. Add product leaving shelf-life empty 2. Save	Field highlighted; error shown; product not saved	
TC-MD-03	FR-33	Negative	Invalid values	Logged in as Manager	1. Enter selling price -5 and temperature range 12 to 8 °C 2. Save	Errors: price must be positive, minimum temperature must be below maximum	
TC-MD-04	FR-33	Positive	Edit threshold	Tomato threshold 10	1. Change threshold to 20 2. Save 3. Set stock to 15	Low-stock alert raised (uses new threshold 20)	
TC-MD-05	FR-33	Positive	Deactivate product	Tomato active with past batches	1. Deactivate Tomato 2. Open batch-entry form	Tomato not selectable for new batches; past batches and reports unchanged	

3. Stock Tracking and FEFO (16 test cases)

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-STK-01	FR-4	Positive	Record received batch	Tomato and Supplier A exist	1. Enter batch: Tomato, Supplier A, 50 kg, received now, expiry today+5 2. Save	Batch saved with unique ID; Tomato stock +50 kg	
TC-STK-02	FR-4	Negative	Expiry before receipt date	Logged in as staff1	1. Enter batch with expiry date earlier than received date 2. Save	Error "Expiry date must be after received date"; not saved	
TC-STK-03	FR-4	Negative	Zero / negative quantity	Logged in as staff1	1. Enter batch quantity 0, then -5 2. Save each	Both rejected with a validation error	
TC-STK-04	FR-5	Positive	Sale taken from earliest-expiry batch	Batch A 10 kg exp 7 Oct; Batch B 20 kg exp 10 Oct	1. Send POS sale of 5 kg Tomato	A = 5 kg, B = 20 kg	
TC-STK-05	FR-5	Boundary	Sale spanning two batches	Batch A 10 kg exp 7 Oct; Batch B 20 kg exp 10 Oct	1. Send POS sale of 15 kg	A = 0 kg, B = 15 kg	
TC-STK-06	FR-5	Negative	Sale above total stock	Total Tomato stock 30 kg	1. Send POS sale of 40 kg	Sale rejected "Insufficient stock"; stock unchanged	
TC-STK-07	FR-6	Positive	Restock adds stock	Banana stock 20 dozen	1. Record restock of 30 dozen	Stock = 50 dozen; restock logged with user and time	
TC-STK-08	FR-8	Positive	Time in store / time left	Batch received 2 days ago, expires in 3 days	1. Open stock screen	Shows "In store: 2 days" and "Expires in: 3 days"	
TC-STK-09	FR-9	Positive	FEFO order suggestion	3 Milk batches expiring 8, 6, 9 Oct	1. Open FEFO suggestion for Milk	Order shown: 6 Oct, 8 Oct, 9 Oct	
TC-STK-10	FR-36	Negative	Sale from expired batch blocked	Only Bread batch expired yesterday	1. Attempt sale of 2 Bread	Sale blocked with "Batch expired"; stock unchanged	
TC-STK-11	FR-36	Boundary	Expired batch skipped	Bread: batch X expired, batch Y valid	1. Sell 2 Bread	Taken from batch Y only; batch X unchanged	
TC-STK-12	FR-37	Positive	Manual sale entry	Logged in as cash1; Tomato 30 kg	1. Enter manual sale 3 kg Tomato 2. Save	Sale recorded; stock deducted in FEFO order	
TC-STK-13	FR-37	Positive	CSV sales upload	Valid CSV with 20 sale rows	1. Upload CSV	All 20 recorded; summary "20 imported, 0 failed"	
TC-STK-14	FR-37	Negative	CSV with bad row	CSV with 19 valid rows + 1 unknown product	1. Upload CSV	Bad row reported with its line number; 19 valid rows imported	

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-STK-15	FR-38	Positive	Stock count adjustment	System stock 50 kg	1. Enter counted qty 47 kg 2. Reason "spillage" 3. Save	Difference -3 kg shown; stock = 47; reason, worker, time saved	
TC-STK-16	FR-38	Negative	Adjustment without reason	System stock 50 kg	1. Enter counted qty 47 kg, no reason 2. Save	Not saved; reason is mandatory	

4. Low-Stock Alerts (5 test cases)

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-LOW-01	FR-10	Positive	Advance alert raised	Threshold 10; stock 15; avg daily sales (7 days) = 6	1. Run daily low-stock check	Alert: "Tomato projected below 10 tomorrow" (15 - 6 = 9)	
TC-LOW-02	FR-10	Negative	No alert when stock is enough	Threshold 10; stock 40; avg 6/day	1. Run daily low-stock check	No advance alert	
TC-LOW-03	FR-10	Positive	Per-item threshold used	Onion threshold set to 25; stock 30; avg 8/day	1. Run daily low-stock check	Alert raised (30 - 8 = 22 < 25); default 10 not used	
TC-LOW-04	FR-11	Positive	Alert when stock falls below threshold	Threshold 10; stock 12	1. Sell 4 units (stock = 8)	Immediate low-stock alert in notification centre	
TC-LOW-05	FR-11	Boundary	Stock exactly at threshold	Threshold 10; stock 12	1. Sell 2 units (stock = 10)	No alert (alert only when below 10)	

5. Expiry, Removal and Markdown (8 test cases)

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-EXP-01	FR-13	Positive	Day-before expiry notification	Paneer batch expires tomorrow	1. Run expiry check	Staff notified "Paneer batch #ID expires tomorrow"	
TC-EXP-02	FR-13	Positive	Expired notification	Paneer batch expires today	1. Run expiry check after expiry time	Staff notified "Expired, remove from shelf"	
TC-EXP-03	FR-15	Positive	Record damaged stock	Tomato batch 20 kg	1. Physically remove 5 kg 2. Mark removal: 5 kg, reason "damaged" 3. Confirm	Stock = 15 kg; worker and time logged	
TC-EXP-04	FR-15	Negative	Removal without reason	Tomato batch 20 kg	1. Record removal 5 kg with no reason 2. Confirm	Rejected; reason mandatory	

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-EXP-05	FR-15	Negative	Removal above batch quantity	Tomato batch 20 kg	1. Record removal of 25 kg	Rejected "Quantity exceeds batch stock"	
TC-EXP-06	FR-42	Positive	Markdown applied by rule	Rule: 20% off at 1 day left; max 40%; Curd price 50	1. Run markdown check on Curd batch with 1 day left	Curd batch price = 40; markdown logged	
TC-EXP-07	FR-42	Negative	Rule above maximum discount	Max discount 40%	1. Manager enters rule of 60% off 2. Save	Rejected "Discount exceeds maximum 40%"	
TC-EXP-08	FR-42	Positive	Markdown sell-through tracked	10 units of Curd marked down	1. Sell 8 marked-down units 2. Open markdown report	Shows 8 of 10 sold (80%)	

6. Refrigeration and Sensor Input (simulated) (7 test cases)

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-REF-01	FR-17	Positive	Unit failure alert	Fridge-2 status OK	1. Simulator sends Fridge-2 status FAILED	Alarm "Fridge-2 failed" on dashboard and notification centre	
TC-REF-02	FR-17	Positive	Recovery shown	Fridge-2 status FAILED	1. Simulator sends Fridge-2 status OK	Fridge-2 shown OK; alarm cleared and logged	
TC-REF-03	FR-23	Positive	Door open too long	Door limit configured as 2 min	1. Simulator sends door OPEN for 3 min	Door-open alarm raised for that unit	
TC-REF-04	FR-23	Boundary	Door open within limit	Door limit 2 min	1. Simulator sends door OPEN, then CLOSED after 1 min	No alarm	
TC-REF-05	FR-43	Positive	Valid reading accepted	Simulator registered with valid key	1. Send temperature reading 4.5 °C for Fridge-1	HTTP 200; reading stored and shown on refrigeration screen	
TC-REF-06	FR-43	Negative	Invalid device key	Wrong or missing key	1. Send reading	HTTP 401; reading not stored; attempt logged	
TC-REF-07	FR-43	Negative	Malformed reading	Valid key	1. Send temperature value "abc"	HTTP 400 with validation message; not stored	

7. Attendance (Low priority / future scope) (5 test cases)

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-ATT-01	FR-24	Positive	Punch-in	Employee E1 not punched in	1. E1 punches in	Punch-in stored with date and time	

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-ATT-02	FR-25	Positive	Punch-out	E1 punched in	1. E1 punches out	Punch-out stored with date and time	
TC-ATT-03	FR-26	Negative	Duplicate punch-in	E1 already punched in	1. E1 punches in again	Rejected "Already punched in"	
TC-ATT-04	FR-26	Positive	Missing punch-out flagged	E1 punched in, no punch-out	1. Run end-of-day check	Missing punch-out flagged for Manager correction	
TC-ATT-05	FR-27	Positive	Daily attendance report	3 employees with punches today	1. Generate daily report	Report lists each employee with in/out times	

8. Reporting and Offline Handling (7 test cases)

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-REP-01	FR-29	Positive	Dashboard values correct	Known test data loaded	1. Open dashboard	Low stock, expiring soon, alarms and stock match database values	
TC-REP-02	FR-29	Positive	Waste percentage	100 kg Tomato received; 10 kg removed in period	1. Open wastage report for that period	Waste = 10%; money lost = 10 kg x cost price	
TC-REP-03	FR-29	Positive	Supplier quality	Supplier A: 50 kg supplied, 5 kg wasted	1. Open supplier quality report	Supplier A waste rate = 10%	
TC-REP-04	FR-29	Positive	Export PDF and CSV	Report open for 1-5 Oct	1. Export as PDF 2. Export as CSV	Both files download; contents match the screen	
TC-REP-05	FR-29	Boundary	Empty date range	No data for chosen range	1. Run report for that range	Message "No data for this period"; no error	
TC-REP-06	FR-30	Positive	Sales buffered offline	App online	1. Disconnect network 2. Record 3 sales 3. Reconnect	All 3 sales synced; no duplicates; stock correct	
TC-REP-07	FR-30	Positive	Sensor readings buffered	Simulator running	1. Cut network for 10 min 2. Restore	All readings synced with original timestamps	

9. Non-Functional Tests (16 test cases)

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-NFR-01	NFR-1	Positive	Stock update time	2 browsers open on stock screen	1. Send 100 sales via simulator 2. Measure time until each shows on 2nd screen	At least 95 of 100 updates appear within 3 s	
TC-NFR-02	NFR-2	Positive	Punch recording time	Attendance enabled	1. Do 20 punches; measure response time	Every punch recorded within 2 s	
TC-NFR-03	NFR-3	Positive	Alert delivery time	Notification recipients set	1. Trigger 20 alerts (low stock, expiry, temperature, unit fault)	Each alert reaches staff within 60 s	
TC-NFR-04	NFR-5	Positive	Load test	5,000 batches loaded; 100 simulated sensors	1. Run Locust with 50 concurrent users for 15 min	No errors; NFR-1 3 s target still met	
TC-NFR-05	NFR-20	Positive	Removal notification time	Notification recipients set	1. Record 10 removals	Each removal notification delivered within 2 min	
TC-NFR-06	NFR-8	Positive	Auto-recovery	System running in Docker	1. Stop backend container 2. Observe	Container restarts automatically; uptime logged over test period	
TC-NFR-07	NFR-9	Positive	30-minute outage, no data loss	Record counts noted	1. Disconnect for 30 min while recording sales and readings 2. Reconnect 3. Compare counts	Zero records lost	
TC-NFR-08	NFR-21	Positive	Backup and restore	Database with test data	1. Run daily backup 2. Restore into a fresh database 3. Compare table counts	Restore succeeds; all counts match	
TC-NFR-09	NFR-10	Positive	Usability with new users	3 people new to the app; 30-min training	1. Ask each to: log a batch, record a sale, record a removal, check a fridge, acknowledge an alert	All tasks completed without help	
TC-NFR-10	NFR-11	Positive	Retention not purged early	Record dated 23 months ago	1. Run data clean-up job	Record still present	
TC-NFR-11	NFR-12	Positive	Temperature records audit-ready	Readings for 7 days	1. Export temperature history	Every reading has unit, value, date/time; no gaps in logged period	
TC-NFR-12	NFR-14	Positive	Passwords hashed	Two users with the same password	1. Inspect users table in database	Only salted hashes stored; the two hashes differ	
TC-NFR-13	NFR-15	Positive	2FA for Manager	2FA enabled for mgr1	1. Log in as mgr1 with correct password 2. Enter wrong code, then correct code	Wrong code rejected; correct code grants access	

TC ID	Req	Type	Title	Precondition / test data	Steps	Expected result	Result (P/F)
TC-NFR-14	NFR-17	Negative	Employee data restricted	Logged in as cash1	1. Request employee details via screen and API	Access denied (403); data stored encrypted in database	
TC-NFR-15	NFR-25	Positive	Store ID on data	Store 1 data exists	1. Add Store 2 and record a batch for it 2. Query as Store 1 user	Works without schema change; Store 1 user sees only Store 1 data	
TC-NFR-16	NFR-26	Positive	Traceability complete	This document	1. Review traceability matrix (Section 4)	Every requirement has at least one test case	

4. Requirements Traceability Matrix

Req ID	Requirement (SRS v1)	Test cases	Count
FR-1	Login with user ID and password	TC-AUTH-01, TC-AUTH-02, TC-AUTH-03	3
FR-2	Role-based access (Admin, Manager, Inventory Staff, Cashier)	TC-AUTH-04, TC-AUTH-05, TC-AUTH-06	3
FR-3	Lockout after 5 failed logins; idle logout	TC-AUTH-07, TC-AUTH-08, TC-AUTH-09	3
FR-4	Record received batch	TC-STK-01, TC-STK-02, TC-STK-03	3
FR-5	Auto-deduct stock on sale in FEFO order	TC-STK-04, TC-STK-05, TC-STK-06	3
FR-6	Add stock on restock	TC-STK-07	1
FR-8	Time in store and time left to expiry	TC-STK-08	1
FR-9	Suggest FEFO selling order	TC-STK-09	1
FR-10	Advance low-stock alert (projection)	TC-LOW-01, TC-LOW-02, TC-LOW-03	3
FR-11	Low-stock alert	TC-LOW-04, TC-LOW-05	2
FR-13	Expiry notifications (day before and on expiry)	TC-EXP-01, TC-EXP-02	2
FR-15	Record non-expiry stock loss after physical removal	TC-EXP-03, TC-EXP-04, TC-EXP-05	3
FR-17	Refrigeration unit failure alert	TC-REF-01, TC-REF-02	2
FR-23	Door-open alarm	TC-REF-03, TC-REF-04	2
FR-24	Attendance punch-in	TC-ATT-01	1
FR-25	Attendance punch-out	TC-ATT-02	1
FR-26	Duplicate punch-in / missing punch-out handling	TC-ATT-03, TC-ATT-04	2
FR-27	Daily attendance report	TC-ATT-05	1
FR-29	Dashboard and exportable reports	TC-REP-01, TC-REP-02, TC-REP-03, TC-REP-04, TC-REP-05	5
FR-30	Offline buffering and sync	TC-REP-06, TC-REP-07	2
FR-31	Admin user management	TC-AUTH-10, TC-AUTH-11, TC-AUTH-12	3
FR-32	Password reset by email	TC-AUTH-13, TC-AUTH-14	2
FR-33	Product master data	TC-MD-01, TC-MD-02, TC-MD-03, TC-MD-04, TC-MD-05	5

Req ID	Requirement (SRS v1)	Test cases	Count
FR-36	Block sale of expired batch	TC-STK-10, TC-STK-11	2
FR-37	Manual and CSV sales entry	TC-STK-12, TC-STK-13, TC-STK-14	3
FR-38	Physical stock count and adjustment	TC-STK-15, TC-STK-16	2
FR-42	Rule-based markdowns	TC-EXP-06, TC-EXP-07, TC-EXP-08	3
FR-43	Authenticated device API for sensor data	TC-REF-05, TC-REF-06, TC-REF-07	3
NFR-1	Stock update on all screens within 3 s (95%)	TC-NFR-01	1
NFR-2	Attendance punch recorded within 2 s	TC-NFR-02	1
NFR-3	Alerts delivered within 60 s	TC-NFR-03	1
NFR-5	50 users, 5,000 SKUs/batches, 100 sensors	TC-NFR-04	1
NFR-8	99.5% availability; alerts on backup power	TC-NFR-06	1
NFR-9	No data loss in 30-min network outage	TC-NFR-07	1
NFR-10	Routine tasks after 30 min training	TC-NFR-09	1
NFR-11	2-year record retention	TC-NFR-10	1
NFR-12	Food-safety (FSSAI) compliant temperature records	TC-NFR-11	1
NFR-14	Passwords stored as salted hashes	TC-NFR-12	1
NFR-15	2FA for Manager and Admin	TC-NFR-13	1
NFR-17	Employee data encrypted, restricted access	TC-NFR-14	1
NFR-20	Removal notification within 2 min	TC-NFR-05	1
NFR-21	Daily backup and tested restore	TC-NFR-08	1
NFR-25	Store ID on all store data	TC-NFR-15	1
NFR-26	Requirement-to-test traceability	TC-NFR-16	1